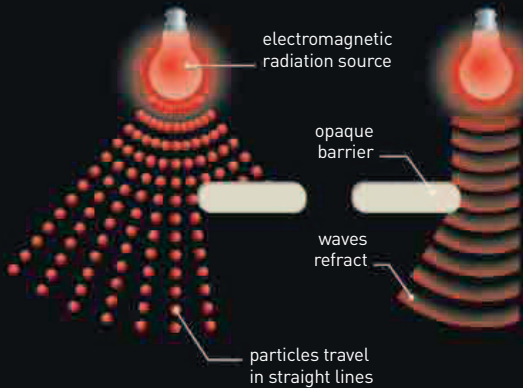


WAVES AND PARTICLES

Scientists had long debated whether light travels through space as streams of particles or as waves. The wave theory was in favor in the 19th century, even before Maxwell's discovery. However, there were phenomena the wave theory could not explain, including the "photoelectric effect."



WAVE-PARTICLE PARADOX

All waves "diffract" or spread out as they pass the edges of stationary objects. Water waves do this as they enter harbors, for example. The diffraction of light is hard to explain if light is understood to be a stream of particles.

In 1887, Hertz attached two electrodes to a battery and set them a small distance apart in a vacuum tube. When he shone light onto them, an electric current fired between them, but above a certain wavelength it stopped, however intense the light. Albert Einstein explained this effect by proving that electromagnetic radiation exists as particles (photons), and that different colors of light, and different forms of radiation, differ in the amount of energy their photons carry.

"X-RAYS WILL PROVE TO BE A HOAX."

William Thomson (Lord Kelvin), British physicist, 1899

USING ELECTROMAGNETIC RADIATION

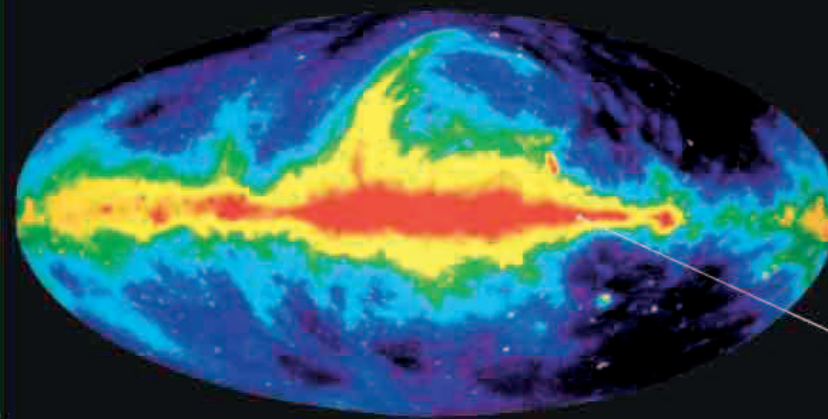
In addition to ultraviolet, infrared, and radio waves, scientists discovered two new forms of electromagnetic radiation in the 1890s, both with very short wavelengths (or high-energy photons): X-rays and gamma rays.

Devices that can produce or detect the various forms of electromagnetic radiation have many, and varied, applications. For example, different types of radio waves are used to carry television, radio, and telephone signals. In medicine, penetrating X-rays are used to produce images of the inside of the body, while gamma rays are used in radiation therapy.



INFRARED RADIATION

Cameras sensitive to infrared radiation can produce color-coded images that reveal temperature differences—allowing engineers to detect heat loss from houses, for example.



RADIO MAP OF THE SKY

Nonvisible electromagnetic radiation provides new windows through which to study the Universe. Normally invisible interstellar dust emits radio waves, so it shows up in this radio map of the sky.

